Semester 1, 2026

Business Plan and Engineering Design Progress Report

EGH419: Advanced Design and Entrepreneurship

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1 Product Name

PowerStake

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Role: Hipster

Primary Responsibility: AI subsystem design, user-focused automation features, and intelligent decision-support systems.

PowerStake is a smart household energy management system designed to reduce electricity costs through appliance-level monitoring, automated recommendations, fault alerts, and a mobile dashboard. The concept combines low-cost sensing hardware with practical analytics to help households reduce standby energy waste, improve appliance efficiency, and simplify energy management decisions.

2 Problem Validation and Engineering Design

2.1 Customer Needs and High-Level Objective

Customer research identified that many households want lower electricity bills but often lack the time, motivation, or technical knowledge to actively manage appliance energy use. Standby loads from televisions, chargers, gaming consoles, printers, and kitchen appliances can create unnecessary ongoing costs. Existing smart plugs commonly provide remote switching or raw power data, however they still rely on users interpreting information and manually taking action.

Customers are more likely to adopt a system that converts power data into clear recommendations, warns of unusual appliance behaviour, and helps reduce wasted energy without increasing user effort. This creates a need for an intelligent analytical subsystem that improves household decision-making through simple and useful outputs.

HLO-004: AI decision engine

The PowerStake AI subsystem must convert appliance-level power monitoring data into actionable fault alerts and energy-saving recommendations that improve household decision-making during prototype testing.

2.2 System Requirements

The following subsystem requirements were developed for the PowerStake AI Decision Engine and are categorised into Physical, Functional, and Performance requirements. While the primary focus is HLO-004, several requirements also support wider system objectives including power measurement, fault detection, automated control, and dashboard usability.

Physical Requirements

REQ-Ph-01
Description: The AI subsystem shall operate on a local prototype computer or cloud-hosted environment.
Rationale: Allows practical prototype deployment using available hardware while supporting

future scalability.
Traced From: HLO-004

REQ-Ph-02
Description: The AI subsystem shall interface with the PowerStake monitoring database and mobile dashboard API.
Rationale: Ensures appliance data can be received and recommendations can be transmitted to end-users.
Traced From: HLO-004, HLO-005

REQ-Ph-03
Description: The AI subsystem shall retain historical appliance usage data for a minimum of 30 days.
Rationale: Provides sufficient behavioural history for trend analysis and anomaly detection.
Traced From: HLO-003, HLO-004

REQ-Ph-04
Description: The subsystem shall support secure wireless communication with the main PowerStake controller.
Rationale: Enables reliable transfer of monitoring data and recommendations between system components.
Traced From: HLO-004, HLO-005

Functional Requirements

REQ-F-01
Description: The AI subsystem shall receive appliance voltage, current, and power data at intervals of 1 second or less.
Rationale: Provides timely data for near real-time analysis and decision generation.
Traced From: HLO-001, HLO-004

REQ-F-02
Description: The AI subsystem shall identify prolonged standby power consumption events.
Rationale: Supports reduction of unnecessary electricity costs caused by vampire power loads.
Traced From: HLO-002, HLO-004

REQ-F-03
Description: The AI subsystem shall detect abnormal appliance behaviour relative to stored historical patterns.
Rationale: Enables early identification of appliance degradation or potential faults.
Traced From: HLO-003, HLO-004

REQ-F-04
Description: The AI subsystem shall generate fault alerts and energy-saving recommendations.
Rationale: Converts raw monitoring data into actionable outputs for the user.
Traced From: HLO-002, HLO-003, HLO-004

REQ-F-05
Description: The AI subsystem shall transmit outputs to the mobile dashboard interface.
Rationale: Ensures recommendations and alerts are visible to the end-user.
Traced From: HLO-004, HLO-005

REQ-F-06
Description: The AI subsystem shall support user override of recommended automated actions.
Rationale: Maintains user control while preserving trust in automated decision-making.
Traced From: HLO-004, HLO-005

Performance Requirements

REQ-Pe-01
Description: Fault alert outputs shall be considered useful in at least 80% of prototype test scenarios.
Rationale: Measures whether generated alerts are relevant and valuable to users.
Traced From: HLO-003, HLO-004

REQ-Pe-02
Description: Energy-saving recommendations shall be considered useful in at least 80% of prototype test scenarios.
Rationale: Validates that recommendations improve household decision-making.
Traced From: HLO-002, HLO-004

REQ-Pe-03
Description: Recommendation outputs shall be generated within 5 seconds of receiving new monitoring data.
Rationale: Maintains responsive near real-time system performance.
Traced From: HLO-004

REQ-Pe-04
Description: The AI subsystem shall process data from at least 10 monitored devices simultaneously.
Rationale: Demonstrates practical household scalability for prototype testing.
Traced From: HLO-004

REQ-Pe-05
Description: Dashboard notifications shall be delivered to the user within 2 seconds of alert generation.
Rationale: Ensures timely communication of faults and recommended actions.
Traced From: HLO-004, HLO-005

2.3 Requirements Traceability Matrix

Table 1 summarises the relationship between subsystem requirements and the PowerStake High-Level Objectives (HLOs). This matrix demonstrates that each requirement supports one or more customer-driven system objectives and provides traceability between stakeholder needs and detailed subsystem design decisions.

Table 1: Requirements traceability matrix for the PowerStake AI subsystem

Requirement ID	**HLO-001**	**HLO-002**	**HLO-003**	**HLO-004**	**HLO-005**
REQ-Ph-01				X	
REQ-Ph-02				X	X
REQ-Ph-03			X	X	
REQ-Ph-04				X	X
REQ-F-01	X			X	
REQ-F-02		X		X	
REQ-F-03			X	X	
REQ-F-04		X	X	X	
REQ-F-05				X	X
REQ-F-06				X	X
REQ-Pe-01			X	X	
REQ-Pe-02		X		X	
REQ-Pe-03				X	
REQ-Pe-04				X	
REQ-Pe-05				X	X

2.4 Subsystem Architecture

The AI subsystem functions as the decision-making and analytics layer of the PowerStake platform. It receives appliance-level monitoring data from the sensing and control subsystems through the residential Wi-Fi communication pathway. Its role is to convert raw electrical measurements into useful outputs including fault alerts, energy-saving recommendations, and suggested automatic shutoff opportunities.

As shown in Figure 1, the subsystem consists of five stages: data ingestion, feature storage, model training, real-time inference, and output delivery. Incoming measurements such as voltage, current, power draw, switching state, and timestamps are processed into useful indicators including standby duration, inactivity periods, usage trends, and abnormal load changes [1].

For the prototype stage, the training layer may use rule-based logic or lightweight machine learning approaches. The inference layer applies the selected model in near real-time to classify appliance behaviour and generate user-facing recommendations [2].

Outputs are delivered to the mobile dashboard, allowing users to review recommendations, receive alerts, and manually override suggested actions. User feedback may also be stored to improve future model performance. This subsystem provides the core intelligence of PowerStake by linking appliance behaviour with practical household energy decisions [1].

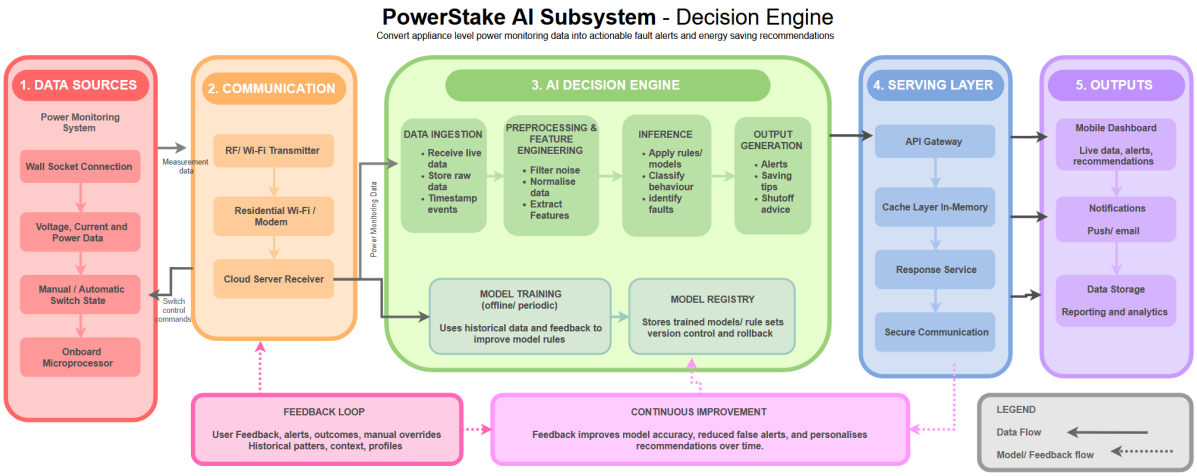

Figure 1: Subsystem architecture for the PowerStake AI subsystem [1].

2.5 Trade Studies

A trade study was conducted to determine the most suitable analytical approach for the PowerStake AI subsystem. The subsystem is required to analyse power monitoring data to identify appliance malfunctions, detect power-saving opportunities, and recommend automatic shutoff actions. For the current prototype stage, the preferred solution needed to be accurate enough to demonstrate value, while remaining practical to implement with limited data, modest computational resources, and a short development timeframe.

Three candidate approaches were evaluated: (1) rule-based thresholding, (2) classical machine learning, and (3) lightweight deep learning. These options were compared based on fault detection capability, data requirement, computational demand, interpretability, and prototype suitability.

Option	**Description**	**Data Requirement**	**Computational Demand**	**Suitability**
Rule-Based Thresholding	Fixed rules using power, current, and inactivity duration [1]	Low	Low	High
Classical Machine Learning	Feature-based classification using engineered inputs [3]	Medium	Medium	Medium
Lightweight Deep Learning	Learns temporal patterns directly from power traces [2]	High	High	Low

Table 2: Initial comparison of AI modelling approaches

Option	Detection	Data Need	Processing	Interpretability	Total
Rule-Based Thresholding	3	5	5	5	18
Classical Machine Learning	4	3	3	3	13
Lightweight Deep Learning	5	2	2	2	11

Table 3: Comparison matrix for AI subsystem options

Rule-based thresholding was selected as the preferred prototype solution. Although it is less sophisticated than machine learning methods, it is the most realistic option for the current project stage because it requires minimal training data, is easy to interpret, and can be implemented rapidly. Classical machine learning is a strong future option once more operating data become available. Lightweight deep learning was considered less suitable at this stage due to higher computational and data requirements. Overall, rule-based thresholding provides the best balance of practicality, transparency, and prototype suitability for demonstrating the AI subsystem value proposition.

References

[1] F. U. Haq. "System design handbook." Ai subsystem design guidelines from the System Design Handbook, Accessed: 2026. [Online]. Available: `https://www.systemdesignhandbook.com/guides/ai-system-design/`.

[2] Google Developers. "Tensorflow lite overview." Framework for deploying lightweight deep learning models on embedded and edge devices, Accessed: Apr. 17, 2026. [Online]. Available: `https://www.tensorflow.org/lite`.

[3] IBM. "What is anomaly detection?" Overview of anomaly detection methods relevant to appliance fault identification, Accessed: Apr. 17, 2026. [Online]. Available: `https://www.ibm.com/topics/anomaly-detection`.